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Multidisciplinary Problem Statements

Common Guidelines:

20-HOUR SCOPE: Prioritize a working Fusion API tool, correct parametric CAD generation, core assembly/motion, and essential validation. Advanced architecture, batch processing and presentation features are not expected.

- Teams may select any ONE of the three problem statements.
 - Autodesk Fusion must be the primary engineering environment. Fusion API must be central to the proposed solution.
 - Python is recommended for API development, but another Fusion-supported API language may be used where appropriate.
 - No starter CAD model will be provided. Participants must create the required CAD geometry themselves from the reference concept/specification.
 - All CAD models created for the solution must be parametric wherever applicable. Key dimensions and relationships must be controlled by parameters rather than unrelated hard-coded geometry.
 - External tools may be used only for supporting calculations, numerical iteration/optimization, data processing, debugging, or similar supporting tasks. Final CAD modelling and Fusion-based engineering operations must remain in Fusion.
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- AI tools may be used for learning, coding assistance, ideation, and debugging. Teams must understand and be able to explain the generated code and engineering decisions.
- Reference images are conceptual guides. The dimensional/specification requirements stated in this document are authoritative.
- Solutions must be reusable. Hard-coding only the exact example answer is not considered a valid automation solution.
- Original work is required. Teams must be able to demonstrate their implementation and explain the logic.



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PROBLEM STATEMENT A

Fusion Configuration Maker

Title: **Parametric Adjustable Roller Conveyor Configuration Generator**

2.1 Objective

Develop a Fusion API tool that automatically generates a configurable roller conveyor module from user-defined requirements. The tool must create and update a parametric CAD assembly through the Fusion API and generate multiple valid configurations by changing the input parameters.

2.2 Reference Concept and Inputs

The reference concept is an adjustable roller conveyor module. Participants must create the model from scratch.

Parameter	Required / Allowed Range	Notes
Conveyor length, L	800–2000 mm	Overall conveyor length
Conveyor width, W	300–600 mm	Overall conveyor width
Frame height, H	500–900 mm	Floor to conveyor frame height
Roller diameter, D	40–80 mm	Roller outside diameter
Roller spacing, P	80–150 mm	Centre-to-centre spacing
Support-leg spacing, S	500–1000 mm	Spacing along conveyor length
Side-guard height, G	0–150 mm	Optional side guard
Side guards	Yes / No	User-selectable feature

2.3 Minimum API Functionality

- Provide a Fusion interface for entering configuration values.
- Create the conveyor geometry and assembly through the Fusion API.
- Generate the frame, rollers, supports, and selected side guards.
- Automatically calculate or update the number and positions of repeated rollers based on the selected length and spacing.
- Automatically position support legs according to the selected conveyor length and support spacing.
- Maintain the selected overall dimensions when parameters are changed.
- Update the parametric model when input values are modified.



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- Generate at least THREE substantially different valid configurations.
- Provide a component list/BOM or equivalent component information for each generated configuration.

2.3A CS/IT Automation Requirements

- Identical inputs should produce the same configuration state.
- Automatically verify key results such as overall dimensions, roller count/spacing and selected features after generation.
- Invalid or conflicting inputs should be rejected before generation where practical.
- Re-running or updating the same configuration should not create obvious duplicate generated components.
- Roller and support-leg counts/positions must be calculated from selected dimensions and spacings, not fixed example layouts.

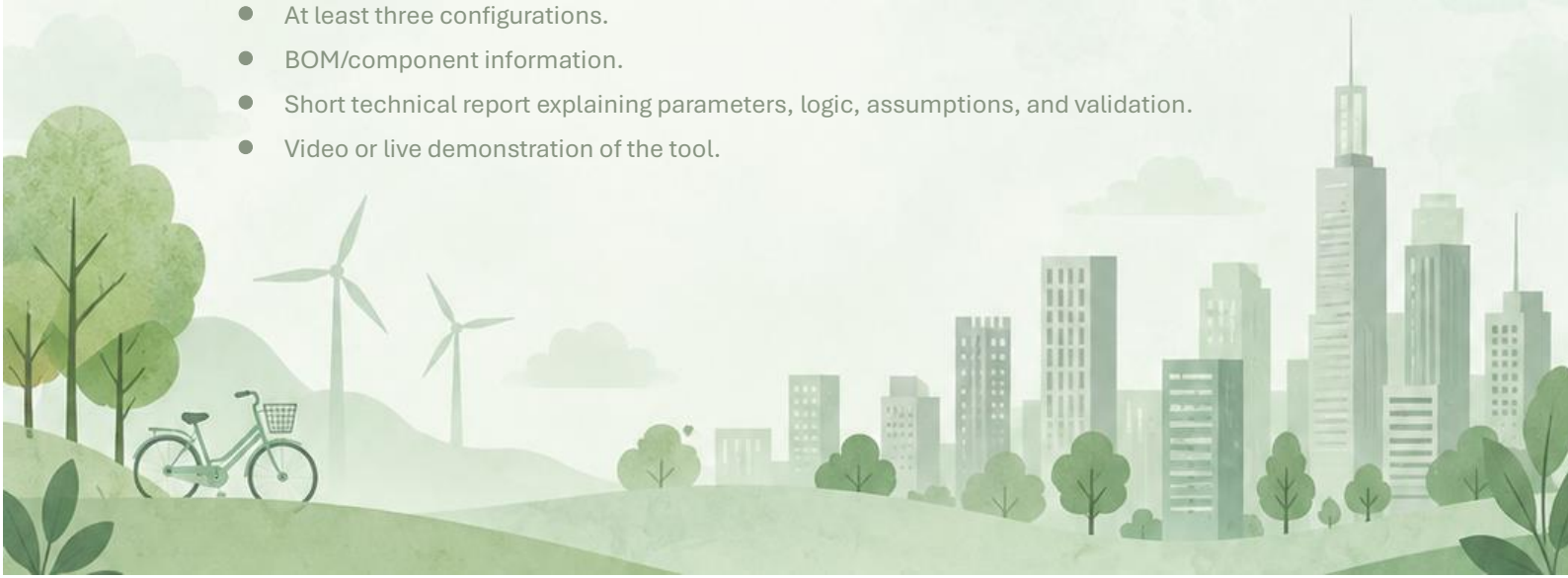
2.4 Validation

- Overall length, width, and height match the entered requirements.
- Roller diameter and spacing satisfy the selected values.
- The number and locations of repeated components update correctly.
- The generated assembly remains coherent after parameter changes.
- The model is genuinely parametric and not a collection of unrelated fixed solids.
- The tool can generate a configuration using inputs other than the demonstration examples.

2.5 Minimum Success Requirement

A valid submission must automatically generate at least THREE different conveyor configurations from changed inputs.

2.6 Deliverables

- Working Fusion API tool.
 - Complete source code.
 - Parametric Fusion model(s) generated by the tool.
 - At least three configurations.
 - BOM/component information.
 - Short technical report explaining parameters, logic, assumptions, and validation.
 - Video or live demonstration of the tool.
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PROBLEM STATEMENT B

Fusion Mechanism Generator

Title: Parametric Mechanical Mechanism Generator

3.1 Objective

Develop a Fusion API tool that automatically generates mechanical mechanisms from user-defined motion requirements. The solution must support at least TWO different mechanism types and create parametric components, joints, and a working motion demonstration.

3.2 Supported Mechanism Types

- Crank-slider mechanism
- Four-bar linkage
- Rocker mechanism
- Cam-and-follower mechanism

Teams may select any two or more of the above mechanisms. A universal generator for every mechanism type is not required.

3.3 Input Requirements

Parameter	Required / Allowed Range	Notes
Input motion	Rotary	Motor/input shaft motion
Input speed	30–120 rpm	Nominal input speed
Reciprocating stroke	50–120 mm	For crank-slider type
Output oscillation angle	30–90°	For rocker/four-bar type
Mechanism dimensions	Defined by selected mechanism	Tool must derive or apply suitable link dimensions
Cam/follower lift	50–120 mm	For cam-and-follower type, where applicable

3.4 Minimum API Functionality

- Provide a Fusion interface for mechanism selection and input values.
- Generate the required component geometry through the Fusion API.
- Create and position components automatically.
- Create appropriate Fusion joints for the selected mechanism.
- Apply the required dimensions parametrically.



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- Demonstrate the generated mechanism through Fusion motion/joint simulation.
- Allow key dimensions or motion requirements to be changed and regenerate/update the mechanism.
- Support at least TWO different mechanism types.

3.4A CS/IT Automation Requirements

- Updating an existing mechanism must not leave obvious duplicate geometry or obsolete joints/components.
- The generator must work for inputs beyond the demonstration examples.
- Include a simple automated verification step that checks the generated mechanism against the requested motion requirement.
- Mechanism dimensions must be calculated from the selected inputs or a documented sizing rule before Fusion geometry is generated.

3.5 Engineering Requirements

- The mechanism must satisfy the specified motion requirement.
- For a crank-slider, the relationship between crank radius and stroke must be respected. For example, a 100 mm stroke requires a 50 mm crank radius.
- For linkage mechanisms, the chosen geometry and assumptions must be explained.
- The generated mechanism should not contain obvious geometric interference under the demonstrated motion.
- The CAD model must remain parametric.

3.6 Minimum Success Requirement

Generate at least TWO different mechanism types, each with a working Fusion assembly and motion demonstration.



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PROBLEM STATEMENT C

Fusion Gear Generator

Title: Parametric Multi-Type Gear Pair Generator

3.1 Objective

Develop a Fusion API tool that generates gear pairs from transmission requirements. The tool must support at least TWO different gear types and generate at least TWO substantially different configurations. The gears must be parametric and correctly positioned for meshing.

3.2 Required Gear Types

- Standard external spur gear
- Standard external helical gear

Teams may implement additional gear types, but the two types above are the minimum requirement.

3.3 Input Parameters

Parameter	Required / Allowed Range	Notes
Gear type	Spur / Helical	At least these two types must be supported
Normal module, mn	1.5–3 mm	Use normal module for spur and helical gears; 1.5–3 mm.
Driver teeth, Z1	18–40	Number of teeth
Driven teeth, Z2	30–80	Number of teeth
Pressure angle	20°	Normal pressure angle, 20°.
Face width, b	6–15 mm	Gear face width
Helix angle, β	0–30°	0° for spur; 10–25° recommended for helical.
Target ratio	Z2/Z1	Transmission ratio

3.4 Required Engineering Relationships

- Transmission ratio: $i = Z2/Z1$.
- For spur gears, pitch diameter: $d = mZ$.
- For helical gears, the team must clearly state the module convention used and apply the corresponding pitch-diameter relationship consistently.

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- The gear pair must be positioned according to the calculated pitch geometry and centre distance.
- The tooth count, gear dimensions, pressure angle, and helix angle must remain consistent with the selected parameters.

3.5 Example Configurations

Configuration	Input	Expected relationship
A – Spur pair	Spur; $m=2$ mm; $Z1=20$; $Z2=40$; $PA=20^\circ$; $b=10$ mm	Ratio 2:1; $d1=40$ mm; $d2=80$ mm; centre distance=60 mm
B – Helical pair	Helical; $m=2$ mm; $Z1=20$; $Z2=60$; $PA=20^\circ$; $b=12$ mm; $\beta=20^\circ$	Ratio 3:1; dimensions and centre distance calculated consistently with the stated helical-gear convention

3.6 Minimum API Functionality

- Provide a Fusion interface for gear type and gear parameters.
- Generate the selected gear geometry through the Fusion API.
- Support at least TWO gear types: spur and helical.
- Generate the required tooth count and parametric geometry from the selected parameters.
- Position the gears automatically for meshing.
- Create an assembly/motion demonstration showing the gear relationship.
- Allow gear type, ratio, and key dimensions to be changed.
- Generate at least TWO different gear configurations with different ratios.

3.6A CS/IT Automation Requirements

- The implementation must work on at least one additional unseen parameter set.
- Re-running the same configuration should not create obvious duplicate gears, joints or stale generated geometry.
- Automatically verify tooth count, pitch diameter, ratio and centre distance against the declared inputs and equations.
- Gear mathematics, geometry construction and verification should be logically separated; a complex software architecture is not required.



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- Tooth geometry must be generated procedurally from the supplied parameters rather than fixed to the demonstration examples.

3.7 Validation

- Gear type matches the selected input.
- Tooth count matches the inputs.
- Pitch geometry and dimensions are consistent with the selected gear type and module convention.
- Transmission ratio matches $Z2/Z1$.
- Centre distance and relative positioning are correct.
- Gears are correctly positioned for meshing and motion.
- The generator works for an additional unseen parameter set.
- The generated gear models remain parametric.

3.8 Minimum Success Requirement

The submission must generate at least TWO gear types and at least TWO different gear-pair configurations. At minimum, one demonstrated configuration must be a spur gear pair and one must be a helical gear pair.

3.9 Deliverables

- Working Fusion API tool.
 - Complete source code.
 - Parametric Fusion gear models and assemblies.
 - At least two gear types and two configurations.
 - Fusion motion/assembly demonstration.
 - Equations, assumptions, and gear-generation logic.
 - Short technical report and demonstration.
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